



One Hand *Choen-ji-in*

National Competition for 2019 ISEF

S e o n g w o n , Y A N G a n d S e o n y o n g , O H



Table of contents

01

Background

- Reasons
- Types of keyboard function

02

Development

- Application idea
- One Hand *Cheon-ji-in*

03

Conclusion

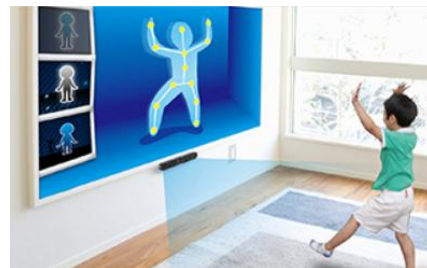
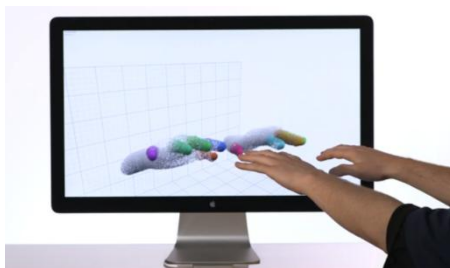
- Conclusion
- Future plan

04

Other works

Arrival of keyboard substitutes

- Recently, traditional keyboard has been challenged by new technologies
- Voice recognition and motion control
- Voice recognition + Artificial intelligence: Siri (Apple), Bixby (Samsung), Alexa (Amazon), NUGU (SKT), Clova (Naver), Cortana (Microsoft), etc Motion control / Wearable devices: LeapMotion, Kinect (color and depth recognition), MYO (muscle recognition), Tap (finger and tapping motion), etc.



구글 홈



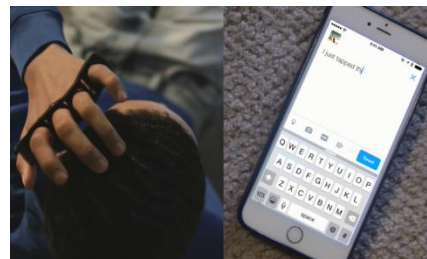
SK텔레콤 누구



네이버 웨이브



카카오 미니



01

Background

Widely adopted voice recognition

- Trendforce, a global market surveyor, estimates that the global voice recognition market size will be USD 16 billion by 2021, five times greater than USD 2.6 billion in 2016.
- The rate of US households that have AI speakers was a meager % in 2016. But that is expected to sharply increase to 5% by 2020. In Korea, the number is estimated to be 1% by 2020.



구글 홈



SK텔레콤 누누

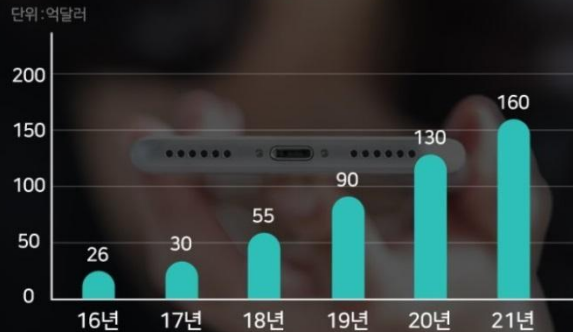


네이버 웨이브



카카오 미니

글로벌 음성인식 시장



출처: 트렌드포스

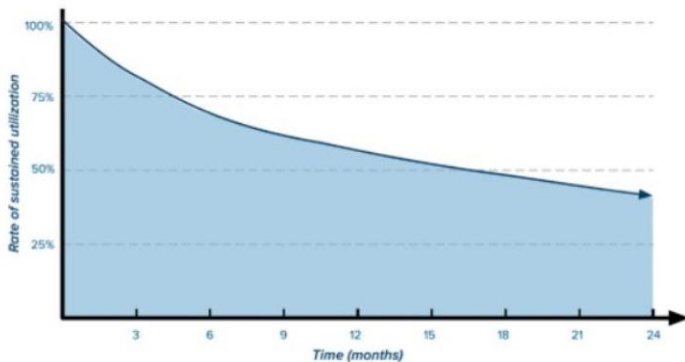
국내 AI 스피커 보유 가구 수 전망



자료: 통계청·한국증권 ※2018년 이후는 추정치

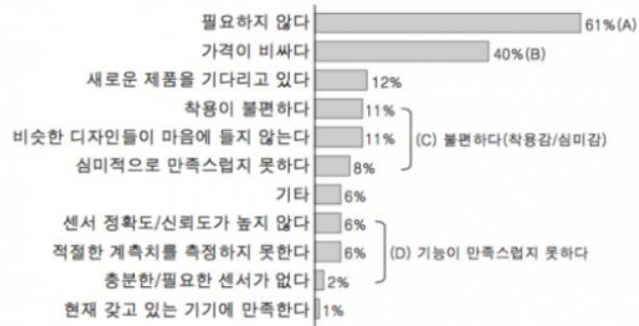
Problems regarding motion control and wearables

- US consulting firm Endeavor Partners said that the rate of sustained utilization of wearables dramatically decreases for the first 6 months of use, ending up with about 50% of utilization after 15 months.
- One of the biggest reasons behind the sharp decrease is that people just want to use what they are used to, rather than an innovative one.



Declining Rate of Sustained Activity Tracker Use Over Ownership
(Endeavour Partners, September 2013)

〈그림〉 웨어러블 기기를 구매하지 않은 이유



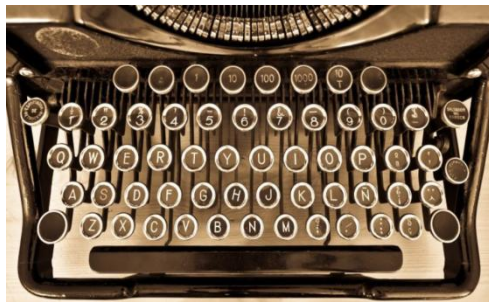
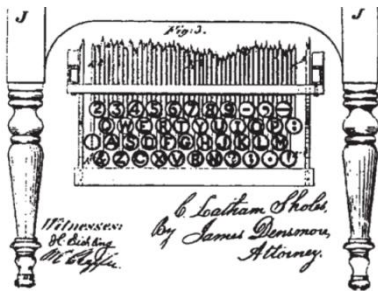
자료 : AlphaWise, 2014(중복 응답)

01

Background

The QWERTY keyboard that has been around over a century

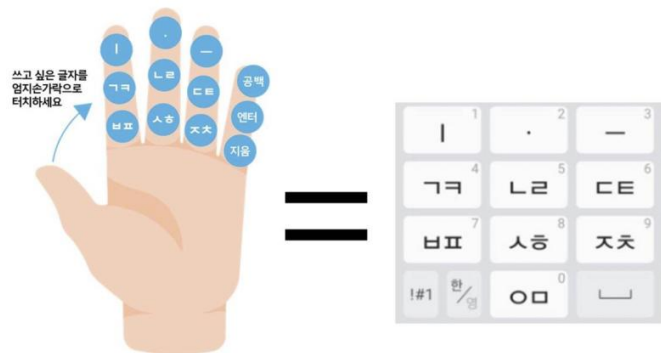
- The QWERTY keyboard has been here with us for over 100 years. Though made in inconvenient ways for people to use, it is still loved by them.
- Indeed, the arrangement system is the most widely used lay-out system not only for English typewriter but computer keyboard. After invented by Christopher Sholes in 1868, It was named after the first 6 letters of your keyboard to the top left. Although some say that efficiency wise it is not so good as Dvorjak Simplified Keyboard, it is the most widely used keyboard arrangement system.



QWERTY keyboard layout
Christopher Sholes, 1878

2019 Present

01 One Hand Choen-ji-in



웨어블 디바이스를 이용해 손가락을
천지인 키보드로 만들었습니다.



02

One-hand Cheon-ji-in

Capacitive Touch Sensor

It senses the change in capacitance, when touched by part of body or certain things, and activates.

It displays High or Low, after identifying the difference between the amount capacitance of surface and that of target, when touched by part of body



Capacitive Touch Sensor Test

Problem

Test result after connecting the sensor to breadboard
-not working properly

Identification

The sensor check the amount of capacitance and display its value.
It does not show correct value when used through breadboard

Solution

Directly connect the sensor through cord
-Activated when its upper part is touched

02

One Hand Cheon-ji-in

Prototype

Wearables should be small in size. To make this happen, Arduino Uno was replaced with Arduino Mini. To check capacitance, cord was placed on fingers other than thumb.



Problem

Case

- Cord is thin that it can easily rolled inside the glove when put on and taken off
- Does not function properly



Solution

Cord was fixed on the outside, without peeling off, and then peel off before fixing



=



웨어블 디바이스를 이용해 손가락을
천지인 키보드로 만들었습니다.

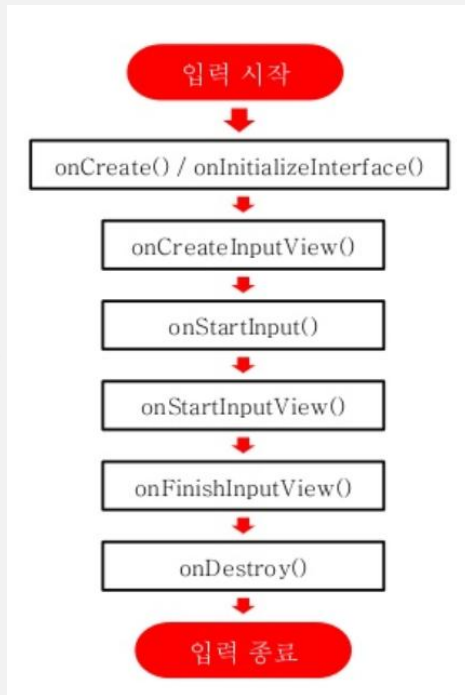
정전용량
터치센서

미니보드

블루투스

어플리
케이션

Keyboard making for Android



One Hand Cheonjiin for Android Keyboard is under development to apply the application program(Kakao Talk, Facebook, or Word Processor) to practical usage.

Improved functionality driven by Big Data based language processing

- The biggest benefit is to type letters without taking a glance at screen
- To commercialize, the accuracy of language input together with voice-recognition technology should be 95% and higher.
- To decrease the error rate of input language done on One Hand Cheon-ji-in and increase the accuracy of the input language, Big-Data based language is employed to improve functionality.

Realization of one hand keyboard language processing interface

- To increase the recognition rate of one hand keyboard, without checking on screen,
 1. Analyzed error patterns of recognition through diverse experiment groups
 2. Needs a huge amount of data to model vocabulary, grammar, context, etc. in order to utilize knowledge in terms of language
- Requires a way to combine one hand keyboard language input frame with a huge amount of Big Data in order to reliable knowledge information statistics.

Realization of language recognition AI using N-gram algorithm

- By using representative language modeling method N-gram, realize entered One hand Cheon-ji-in input in an appropriate grammatical order without looking at screen.
- N-gram method: n-gram is a contiguous sequence of n items from a given sample of text or speech.
- To produce more elaborate N-gram not only with various corpus but also with modeling patterns of error made by actual users

Beacon based AI Carrier

Beacon

- able to receive different locational information through smartphone while moving
- able to send data to the user without complex connection
- able to send data without user intervention
- Decrease battery consumption and increase useful life due to low power system
- able to send data in broader scopes

Made an application that can communicate with smartphones to receive signal's strength from firmware

Attached beacon at each vertexes of rectangular carrier using 4 beacons. Designed to follow the direction of the nearest beacon that sends out the strongest signal, by comparing among them in every seconds.

Making a carrier and selecting a place for washing device

- Add sprinkler with cleaning function

Fine dust sensor and electromagnetic wave

[MAKER club: Chosen as a excellent club]

- Fine dust sensor takes in the air nearby an air-intake and measures fine dust concentration by using ultra-red LED and light sensor.
- An electric field forms near capacitance, and magnetic field shapes charge nearby current, the flow of charge. All of these makes charge vibrate, producing elect ro-magnetic waves.

Using the above principle, isolated membrane of different polarity makes fine dust move in between them. This is how the fine-dust-proof net was invented.

Thank you.